로봇공학

OpenCV를 이용한 가벼운 사물인식

국립경국대학교 로봇공학전공

이상헌

작업환경 만들기

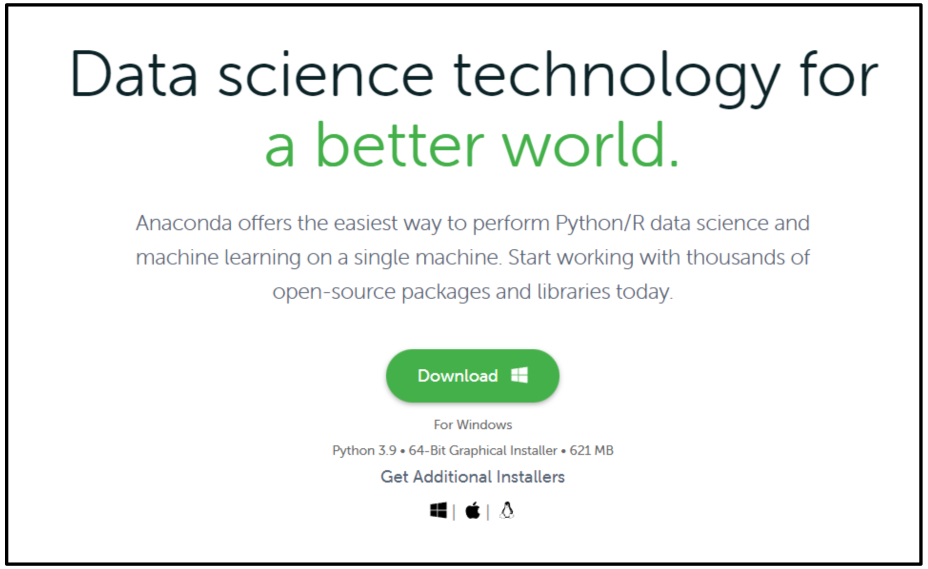
2025.8월 기준

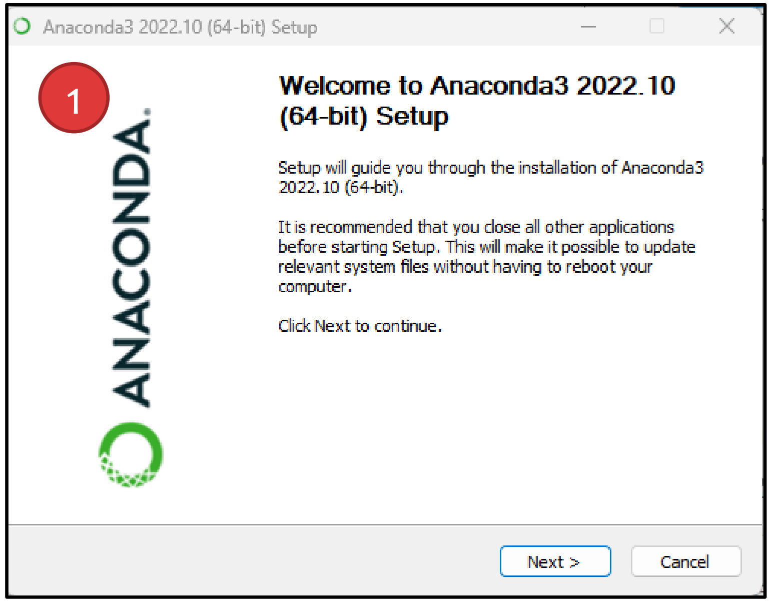
환경 만들기

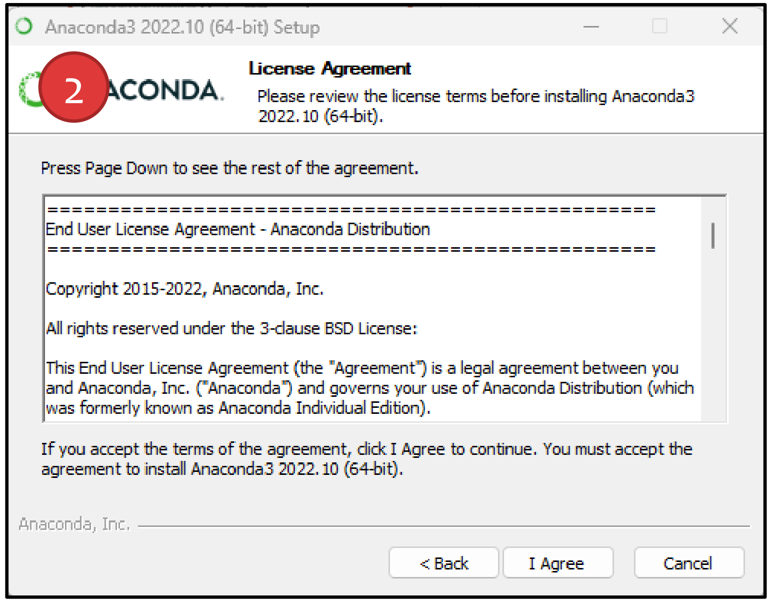
- Python IDE :

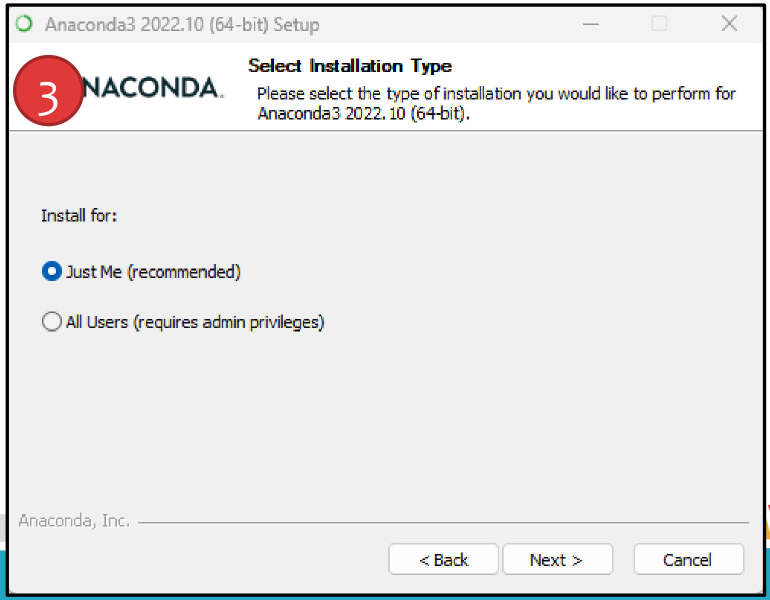
1. **Anaconda3 또는 miniconda**
   - Python 3.10
   - Visual studio code
   - openCV

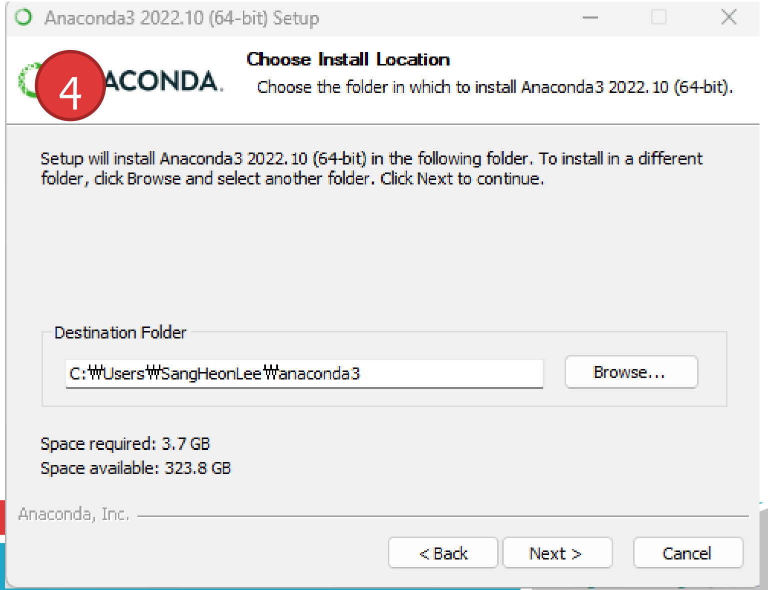
Anaconda Install



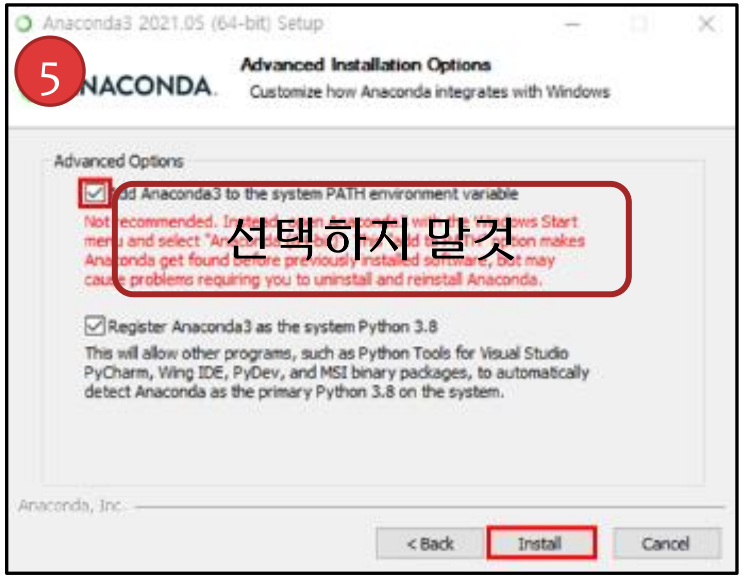


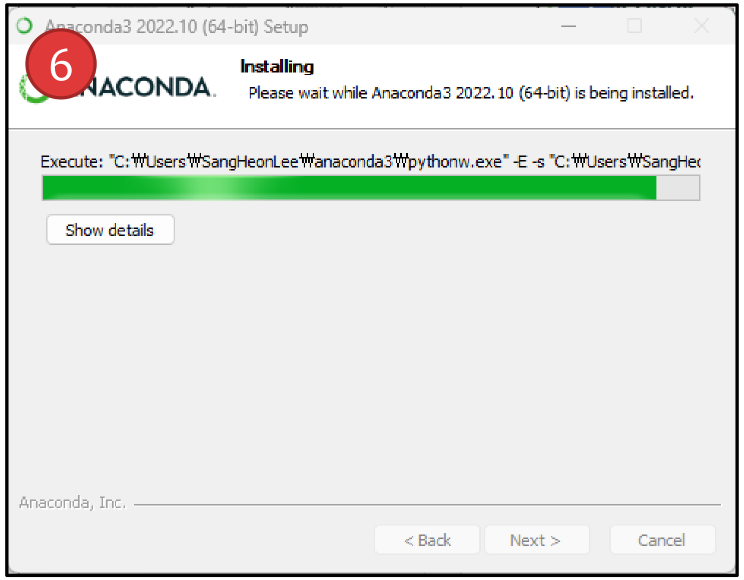


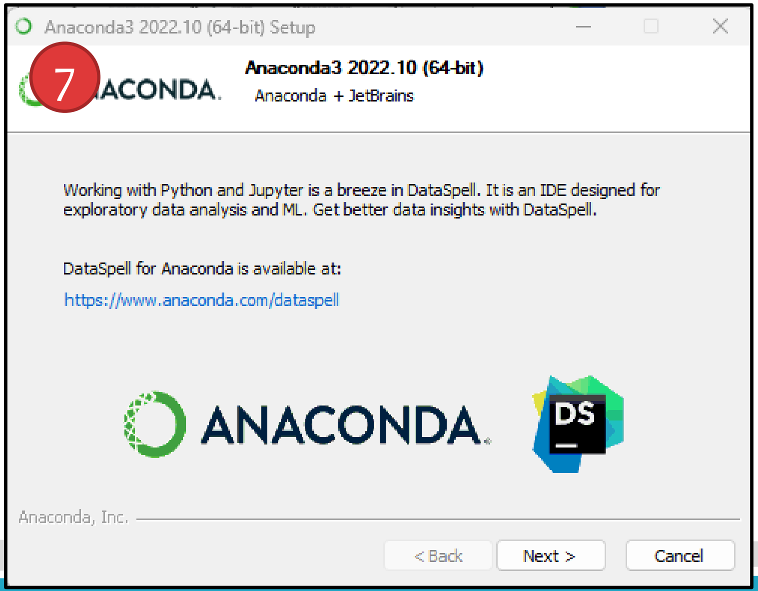


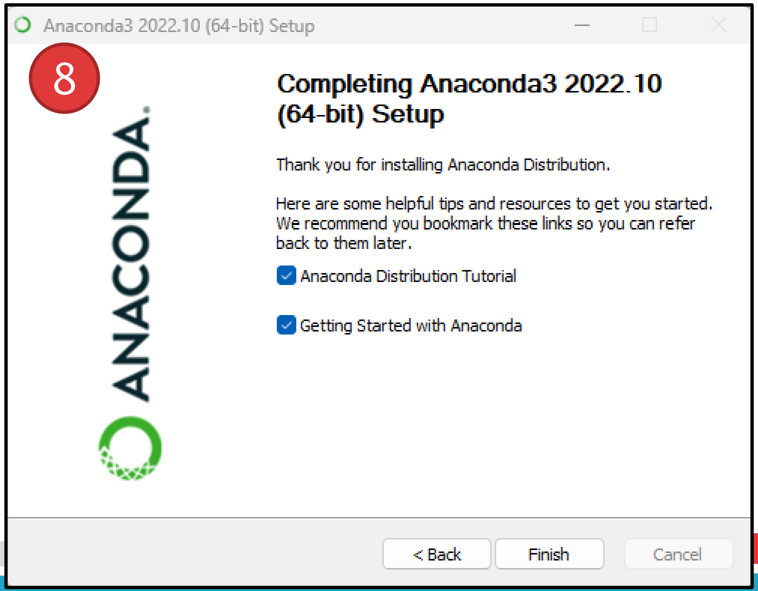


Anaconda Install

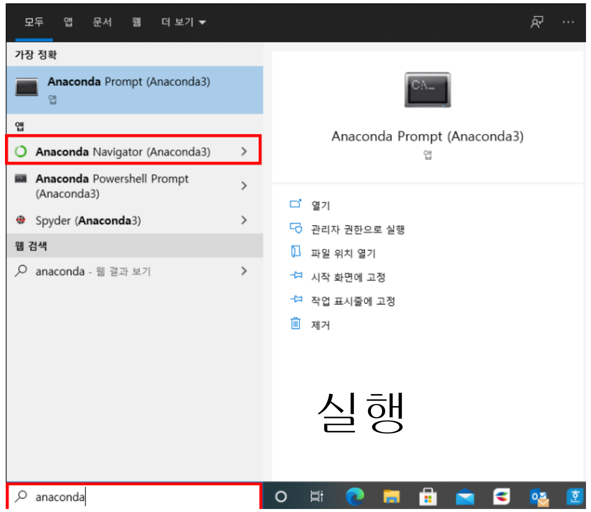




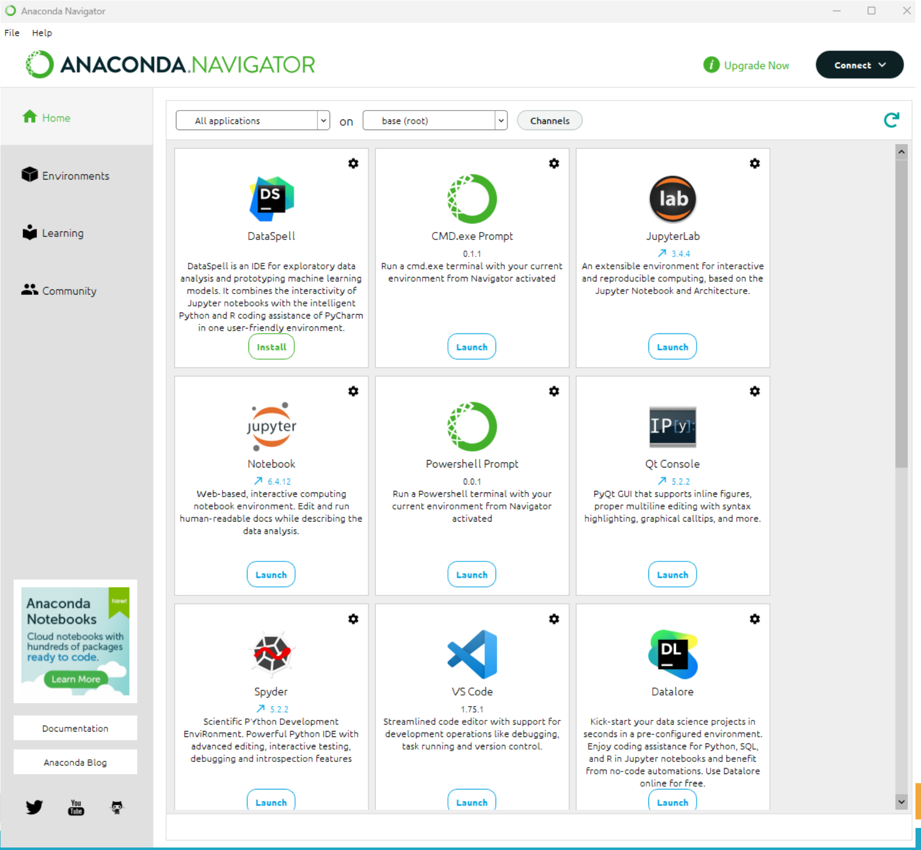


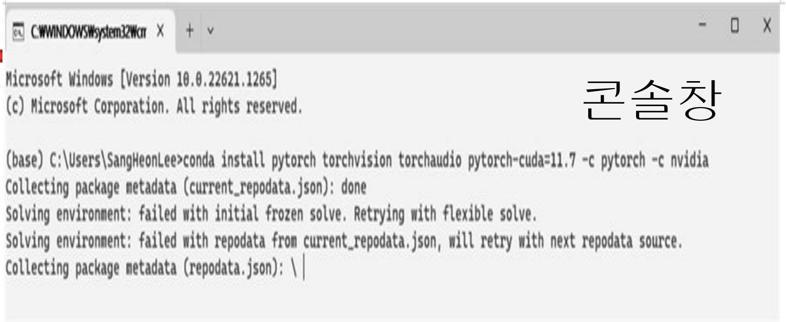


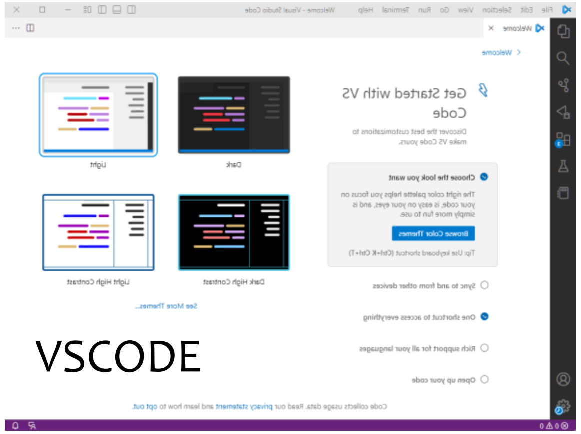
Anaconda 실행

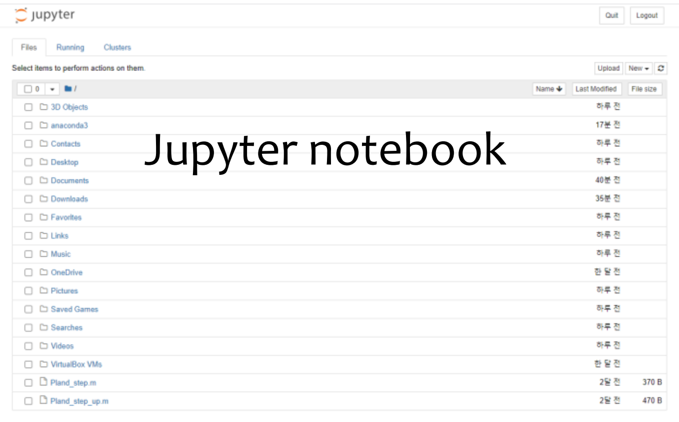


네비게이터









Anaconda 가상환경

가상환경 명령

- conda info –e
- conda create –n yolov5 python=3.9.12
- conda activate yolov5
- conda deactivate
- conda env list
- conda list
- conda env remove –n [ev]

가상 작업환경

1. 가상환경 생성
% conda create -n robot25 python=3.10

2. 가상환경 활성화
% conda activate robot25

3. OpenCV 설치
%%pip install opencv-python numpy

%: base
%%: robot25

AI 모델 파일 다운로드 (MobileNet-SSD)

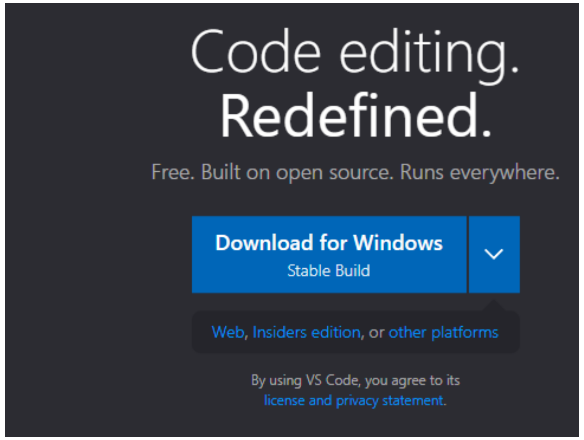
- 🔗 deploy.prototxt
https://github.com/chuanqi305/MobileNet-SSD/blob/master/deploy.prototxt

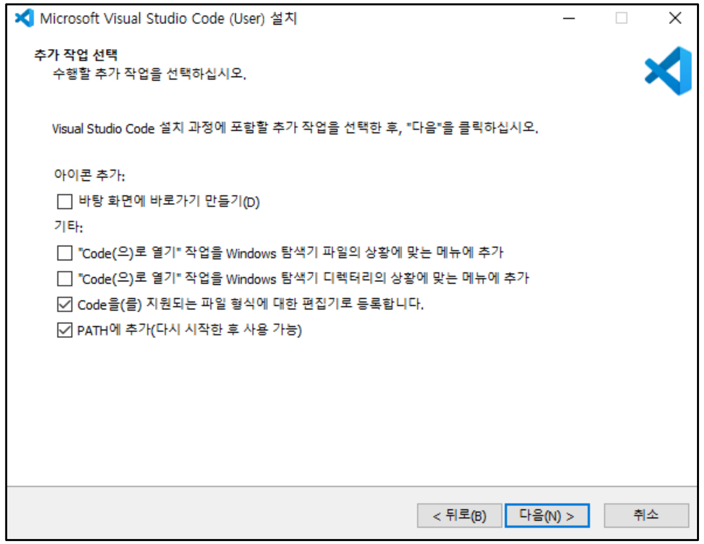
- 🔗 mobilenet_iter_73000.caffemodel
https://github.com/chuanqi305/MobileNet-SSD/raw/master/mobilenet_iter_73000.caffemodel

웹에서 "Raw" 또는 " Download"로 직접 작업홀더로 다운로드 하기

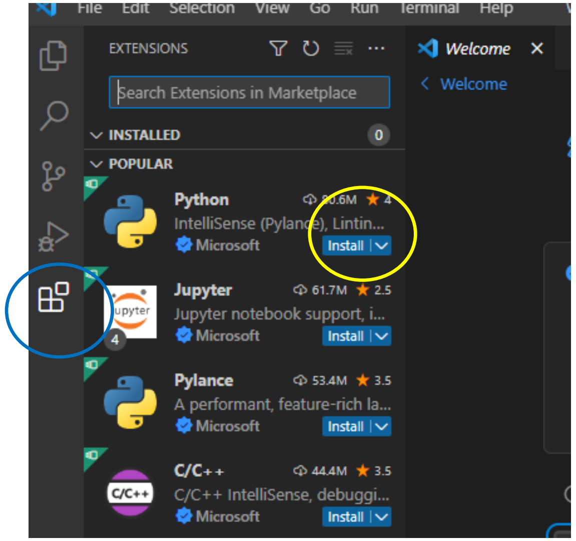
Visual Studio Code Install https://code.visualstudio.com/

1. 다운로드후 인스톨

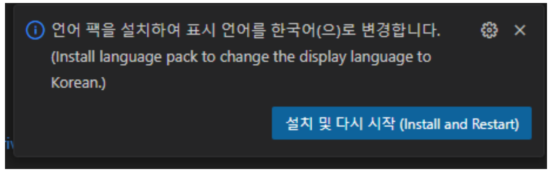




2. Python extension 설치

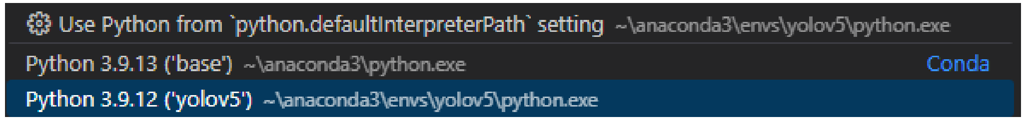


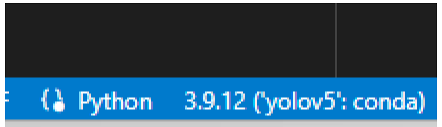
한국어설치 X



3. Python interpreter 변경

- Ctrl+Shift+P
- Select interpreter 입력 후 엔터
- Python 3.10(robot25)





VSCode 실행
%%code

작업홀더 설정하기

예제 코드-1

```
import cv2
import numpy as np

# 클래스 이름 (VOC 20종)
CLASSES = ["background", "aeroplane", "bicycle", "bird", "boat",
        "bottle", "bus", "car", "cat", "chair", "cow", "diningtable",
        "dog", "horse", "motorbike", "person", "pottedplant", "sheep",
        "sofa", "train", "tvmonitor"]

# 모델 파일 경로
prototxt_path = "deploy.prototxt"
model_path = "mobilenet_iter_73000.caffemodel"

# 모델 로딩
net = cv2.dnn.readNetFromCaffe(prototxt_path, model_path)

# 웹캠 열기
cap = cv2.VideoCapture(0)

print("Starting object detection. Press 'q' to quit.")
```

```
while True:
    ret, frame = cap.read()
    if not ret:
        break

    h, w = frame.shape[:2]

    # DNN 입력 전처리
    blob = cv2.dnn.blobFromImage(frame, scalefactor=0.007843,
                                 size=(300, 300), mean=127.5)
    net.setInput(blob)
    detections = net.forward()
```

예제 코드-2

```
# 탐지 결과 처리
  for i in range(detections.shape[2]):
    confidence = detections[0, 0, i, 2]

    if confidence > 0.5:
      idx = int(detections[0, 0, i, 1])
      box = detections[0, 0, i, 3:7] * np.array([w, h, w, h])
      (startX, startY, endX, endY) = box.astype("int")

      centerX = int((startX + endX) / 2)
      centerY = int((startY + endY) / 2)

      label = f"{CLASSES[idx]}: {confidence:.2f}"
      print(f"[{i}] {label}, Center: ({centerX}, {centerY})")

      # 시각화
      cv2.rectangle(frame, (startX, startY), (endX, endY), (0, 255, 0), 2)
      cv2.circle(frame, (centerX, centerY), 5, (0, 0, 255), -1)
      cv2.putText(frame, label, (startX, startY - 10),
              cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 2)
```

```
 cv2.imshow("Object Detection", frame)

   # 종료 키
   if cv2.waitKey(1) & 0xFF == ord('q'):
     break

# 정리
cap.release()
cv2.destroyAllWindows()
```